

Aniruddh Herle

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Research Profile

Cosmologist working at the interface of large-scale structure, intrinsic alignments, and hydrodynamical simulations. My research combines cosmological simulations ([FLAMINGO](#), [COLIBRE](#)), weak lensing statistics, and statistical methodology to understand galaxy–halo connections and systematic effects for next-generation surveys such as *Euclid*, LSST and Roman. I develop robust two-point statistical estimators and halo shape measurement techniques with direct application to precision cosmology.

Current Position

2023–2027 **PhD Candidate in Cosmology**, Leiden Observatory, The Netherlands
Advisors: Prof. Henk Hoekstra (Leiden University), Dr. Elisa Chisari (Utrecht University)
Title: Intrinsic alignments in hydro-dynamical simulations. Research focus: intrinsic alignments, halo shapes, weak lensing statistics, hydrodynamical simulations, galaxy–halo connection.

Education

2021–2023 **MSc Astrophysics**, Ludwig Maximilian University of Munich, Germany
Grade: 1.48 (Distinction in German system)
Thesis: Selection functions of strong lens finding neural networks ([publication](#))
Advisors: Dr. Simona Vegetti (Max Planck Institute for Astrophysics), Prof. Dr. Daniel Gruen (University Observatory of Munich)

2017–2021 **B.E. Electrical and Electronics Engineering**, M. S. Ramaiah Institute of Technology, India
CGPA: 9.38/10 (Distinction)

First Author Publications

- 2026 **A. Herle**, N. E. Chisari, H. Hoekstra, D. Neumann
“[Assembly bias and the redshift evolution of intrinsic alignments for LRGs](#)”; *Astronomy and Astrophysics*.
Analysis of redshift evolution of the alignment signal in hydro-dynamic simulations, with recommendations for observational modeling for *Euclid*, LSST and Roman, as well as the first analysis of assembly bias on IA.
- 2026 **A. Herle**, N. E. Chisari, H. Hoekstra et al.
“[Intrinsic alignments in the FLAMINGO simulations with two-point statistics](#)”; *Astronomy and Astrophysics*.
Comprehensive analysis of intrinsic alignment modelling in next-generation hydrodynamical simulations with the introduction of the TATT-M model, which will be the fiducial alignment model for *Euclid*
- 2025 **A. Herle**, N. E. Chisari, H. Hoekstra et al.
“[Unbiased estimates of the shapes of haloes using the positions of satellite galaxies](#)”; *Astronomy and Astrophysics*.
Introduced a statistically unbiased halo shape estimator robust to satellite sampling limitations.
- 2023 **A. Herle**, C. M. O’Riordan, S. Vegetti
“[Selection functions of strong lens finding neural networks](#)”; *Monthly Notices of the Royal Astronomical Society*.
Quantified detection biases in machine-learning-based strong lens searches.
- 2020–2022 A total of 4 First-author publications and several co-authored publications on machine learning applications to quasar detection, Li-ion battery modelling, and federated learning. Published in IEEE conferences and the *International Journal of Energy Research*. Full list available on [Google Scholar](#).

In-preparation and Co-author Publications

- 2026 **A. Herle**, P. Vielzuf, N. E. Chisari, H. Hoekstra et al.
“Can’t aVOID IA: intrinsic alignments of galaxies around voids” (*in prep.*)
- 2026 A. Halder, **A. Herle**, K. Lehman et al.
“Halo inertia tensors as random matrices” (*in prep.*)
- 2023 Rojas et al.
“The impact of human expert visual inspection on the discovery of strong gravitational lenses” *Monthly Notices of the Royal Astronomical Society*.

Expert review

Served as expert referee for ApJ, *Euclid* internal review and Royal Astronomical Society Techniques and Instruments.

Contributed Talks at international conferences

- 2026 Intrinsic alignment workshop, IAP, Paris, France
- 2025 NOVA Network 1 Cosmology Meeting, Leiden University, Leiden, the Netherlands
Dark Universe Science Collaboration Meeting, University of Groningen, Groningen, the Netherlands
echo-IA Meeting, Durham, Duke University, USA
Lunch Talk, Observatory of Munich, Munich, Germany
Lunch Talk, University of Strasbourg, Strasbourg, France
- 2024 MOCK Barcelona, INAF, Barcelona, Spain ICTP Summer School, ICTP, Trieste, Italy
- 2023 Machine Learning in Astronomy, Paris Observatory, Paris, France

Supervision

- 2026 **Primary supervisor**, MSc thesis – Kostas Akalestos
Project: Dark matter halo shape recovery in mock Euclid observations.
- 2025 **Primary supervisor**, MSc thesis – Lea Krarup
Project: Substructure detection in weak lensing maps of galaxy clusters.

Teaching Experience

- 2024–2027 Teaching Assistant, “Large Scale Structure and Galaxy Formation”
Instructor: Prof. Dr. Joop Schaye

Technical Expertise

- Hydrodynamical simulations: FLAMINGO, COLIBRE
- Advanced statistical analysis of large-scale structure and intrinsic alignments
- Two-point statistics and correlation function estimators
- High-performance computing (SLURM clusters, GPU acceleration)
- Python scientific stack with a decade of experience (NumPy, SciPy, Astropy, PyTorch)
- Machine learning for astrophysical data analysis

Service and Outreach

- 2026 **Local Organising Committee (LOC)**, UtrechIA echo-IA Annual Meeting (2026).
- 2024-2025 **Core member, Social Committee** -co-organiser of weekly gatherings for fellow colleagues.
- 2023-2027 **Core member, Astronomy on Tap Leiden** – co-organiser of monthly public outreach events.